

Homework (Habanero)

Q1. Simplify $\sqrt{64}$

- (A) 8
- (B) 16
- (C) 32
- (D) 4
- (E) 6

Q2. Convert $\sqrt[3]{27}$ to rational exponent form

- (A) $3^{\frac{1}{3}}$
- (B) $3^{\frac{2}{3}}$
- (C) $3^{\frac{3}{3}}$
- (D) $3^{\frac{4}{3}}$
- (E) $3^{\frac{5}{3}}$

Q3. A patient is prescribed $2 \cdot \sqrt{5}$ milligrams of medication. If $\sqrt{5} \approx 2.24$, how many milligrams of medication will the patient take?

- (A) 4.48
- (B) 5.28
- (C) 6.24
- (D) 7.2
- (E) 8.16

Q4. Simplify $\sqrt{16x^2}$

- (A) $4x$
- (B) $4|x|$
- (C) $16x$
- (D) $16|x|$
- (E) $8x$

Q5. The volume of a cell is increasing at a rate of $2 \cdot \sqrt{t}$ cubic millimeters per hour. If t is the number of hours, what is

the volume of the cell after 4 hours?

- (A) $4 \cdot \sqrt{4}$
- (B) $8 \cdot \sqrt{2}$
- (C) 16
- (D) $4 \cdot 2$
- (E) 8

Q6. Convert $x^{\frac{2}{3}}$ to radical form

- (A) $\sqrt[3]{x^2}$
- (B) $\sqrt[2]{x^3}$
- (C) $\sqrt[3]{x}$
- (D) $\sqrt{x}$
- (E) $\sqrt[2]{x}$

Q7. The probability of a patient responding to a new medication is $\frac{1}{\sqrt{2}}$. What is the probability that the patient will not respond to the medication?

- (A) $\frac{1}{2}$
- (B) $\frac{1}{\sqrt{2}}$
- (C) $1 - \frac{1}{\sqrt{2}}$
- (D) $\frac{1}{2}$
- (E) $1 - \frac{1}{2}$

Q8. A scientist is studying the growth of a cell culture. The number of cells is increasing at a rate of $3 \cdot \sqrt{t}$ cells per hour. If t is the number of hours, what is the number of cells after 9 hours?

- (A) $9 \cdot 3$
- (B) 27
- (C) $3 \cdot 3$
- (D) $3 \cdot \sqrt{9}$
- (E) $27 \cdot 3$

Open Ended Questions (FRQ)

Q9. The length of a microscope slide is $2 \cdot \sqrt{5}$ centimeters. If the width of the slide is x centimeters, and the area of the slide is $20\sqrt{5}$ square centimeters, find the value of x . Show all work and simplify your answer.

Q10. The volume of a cell is given by the formula $V = \frac{4}{3}\pi r^3$, where r is the radius of the cell. If the radius of the cell is increasing at a rate of $2 \cdot \sqrt{t}$ millimeters per hour, find the rate at which the volume of the cell is increasing after 4 hours. Show all work and simplify your answer.

Chef's Tips

- Ensure students show all work and simplify their answers.
- Emphasize the importance of checking units in word problems.
- Encourage students to use calculators to check their work, but not to rely solely on them.